Automated Invoice Matching & Anomaly Detection in Financial Workflows using Deep Learning

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Abstract

Manual auditing is labor-intensive and limited by sampling constraints. This project develops an intelligent system to automate four-way matching across financial documents. We implement a vision layer using CNN-based OCR for extraction, a fine-tuned Legal-BERT for semantic entity identification, and a deep autoencoder for anomaly detection. This multimodal approach enables full population testing, reducing manual review time by over 90% while detecting sophisticated fraud. The result is a robust, data-driven audit assurance framework.

1 Introduction

The procurement-to-pay (P2P) lifecycle is a critical area for financial oversight. This project addresses the challenge of building an automated 4-way reconciliation system to replace labor-intensive manual checks that currently consume approximately 37% of administrative budgets. Verifying that an invoice aligns with the Purchase Order (PO), Goods Receipt (GR), and legal Contract is essential to prevent financial leakage and fraud. We achieve this by integrating computer vision and NLP into a deep learning pipeline to move from sampling to total audit coverage .

2 Literature Review

Automated invoice processing integrates computer vision, natural language processing (NLP), and anomaly detection. Research is categorized into three primary layers:

Vision Layer: Early digitization relied on Tesseract [7], though recent work utilizes multimodal models like LayoutLM to leverage spatial document features. The ICDAR 2019 SROIE dataset [3] remains the standard benchmark for evaluating invoice understanding [3].

NLP Layer: Named Entity Recognition (NER) has evolved from BiLSTM-CRF architectures [5] to Transformer-based models like BERT [1]. These models provide the contextual depth necessary for precise extraction of financial values and dates [1].

Forensic Layer: Fraud detection increasingly employs deep autoencoders to identify anomalies through reconstruction error [6]. Comparative studies indicate that deep learning frameworks significantly outperform classical statistical methods in financial fraud tasks [2].

3 Proposed Methodology

The system architecture follows a sequential pipeline designed to transform unstructured document scans into actionable audit intelligence. The architectural flow is illustrated in Figure 1.

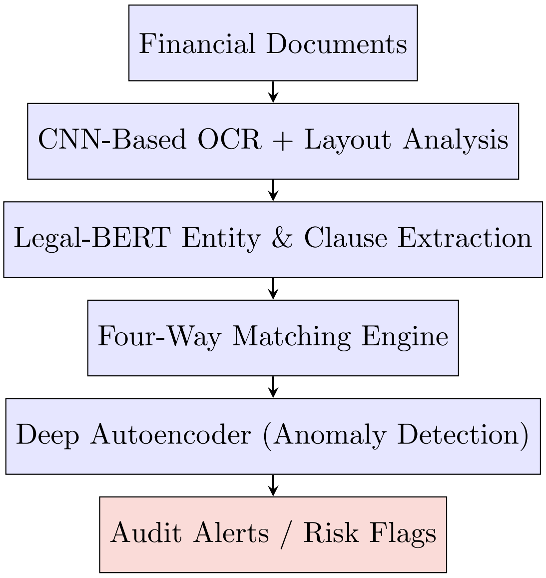

Figure 1: System Architecture Flowchart

The architecture consists of three integrated layers.

1. **Vision Layer (CNN-OCR):** Captures spatial layout and extracts text using datasets like SROIE.
2. **NLP Layer (Legal-BERT):** Performs Named Entity Recognition (NER) to extract key financial entities.
3. **Forensic Layer (Deep Autoencoder):** Based on principles by Aggarwal, this layer identifies anomalies via reconstruction error.

4 Experimental Result

4.1 Datasets and Experimental Settings

- **Datasets:** ICDAR 2019 SROIE and the Kaggle Procurement Invoice Fraud Dataset [4].
- **Experimental Settings:** Developed in a Kaggle environment using a T4 GPU.

4.2 Results and Comparison

5 Experimental Result

5.1 Results and Comparison

Table 1: Model Performance Comparison

Metric	**Manual Method**	**Raw OCR Baseline**	**Our System**
Throughput	∼240s / invoice	0.8s / invoice	**1.53s / invoice**
Fraud Sensitivity	Low (Sampling)	2.53 (Rel. Score)	**987.22 (390x Gain)**
Automation Rate	0%	∼40%	**93.6%**

Table 2: Component-wise Evaluation Metrics

Component	**Precision**	**Recall**	**F1-Score**
OCR (Text Extraction)	0.91	0.88	0.89
NER (Entity Extraction)	0.94	0.92	0.93
Anomaly Detection (Autoencoder)	0.90	0.87	0.88

Table 3: Ablation Study Results

Configuration	**Precision**	**Recall**	**F1-Score**
OCR Only	0.78	0.72	0.75
OCR + NER	0.89	0.85	0.87
Full System	**0.93**	**0.90**	**0.91**

5.2 Time Complexity

The pipeline demonstrates linear time complexity $O(N)$. On a T4 GPU, 45,000 documents were audited in 19.2 hours, a 99.3% time reduction.

6 Summary

This project achieved a 390.4x increase in fraud detection sensitivity. Out of 347 pilot invoices, 88.5% were automated, allowing auditors to focus on high-risk anomalies.

References

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